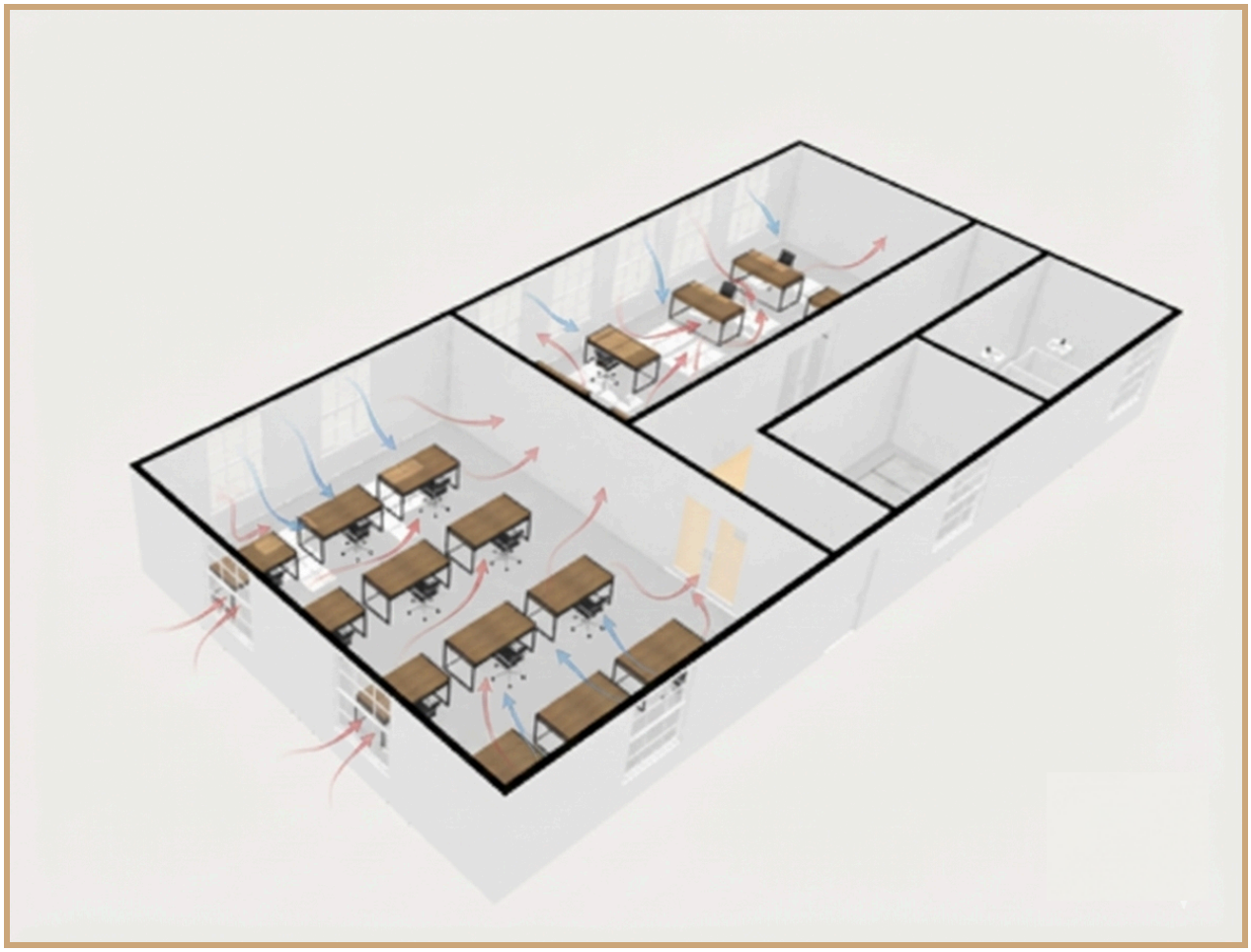
Carlos Mollá

Application to an Office Building in Palermo, Italy

MECA0034-Energy Flexibility in Buildings



LIÈGE université

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Acronyms

HVAC: Heating, Ventilation and Air Conditioning

AHU: Air Handling Unit

HP: Heat Pump

TMY: Typical Meteorological Year

SF: Shading Factor

Introduction

The objective of this project is to design a strategy that moves towards a zero-energy building while maintaining acceptable indoor thermal comfort. This report documents the simulation and analysis of an office building located in Palermo, Italy, using the reference report : *D'Agostino et al., 2019.*

A sequence of simulations will be followed to evaluate and improve the building´s flexibility:

1. Building Model: A simulation of the building without active systems will be done to determine the initial point of the temperatures of the building and the heating and cooling loads required to acclimatize the building.
2. Ventilation System: An Air Handling Unit (AHU) is modeled to quantify the heating and cooling loads provided. Additionally, a specific setpoint strategy is defined to determine the ventilation supply temperature throughout the year.
3. Active Energy Systems: Finally, Terminal Units will be integrated to handle the remaining sensible loads.
4. Generation system: All the component energy will be provided by a generation system consisting of a heat pump ,operating as a chiller in summer.

Project Definition & Boundary Conditions

Site Location

The project is located in Palermo, Italy (38.11° N, 13.36° E). The city has a climate characterized by moderate seasonality, with soft winters and hot summers. However, humidity levels remain significant throughout the year and dehumidification will be necessary.

Climate Analysis

The simulation will use TMY data to represent standard weather conditions. The following table summarizes key parameters of the climate extracted from the TMY data:

Parameter	Metric	Value
Dry Bulb Temperature	Maximum	33.52ºC
	Minimum	2.12ºC
	Average	**17ºC**
Relative Humidity (RH)	Maximum	100%
	Minimum	16.65%
	Average	**74.66%**

Table 1: Key climate parameters

Although Palermo has a warm climate and the temperature range is not particularly demanding for the HVAC system, the high average relative humidity of 74.66% suggests that dehumidification will have an important role in the system. This means the AHU will need to remove a significant amount of water vapour to keep indoor humidity in the comfort range.

Building Typology and Occupancy Schedule

The project simulates an office building with a total area of 200 m^2. The building is divided into five distinct zones, which are shown in the figure below:

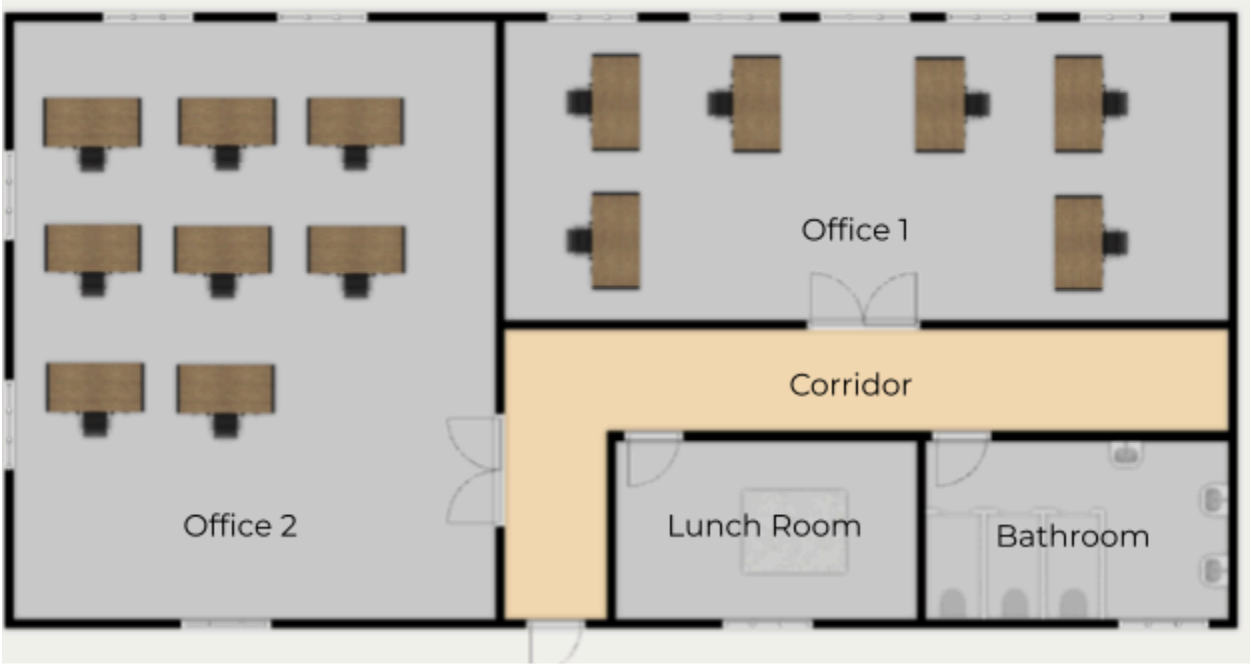


Figure 1: Architectural layout and distribution of the office level

Zone	Zone surface m²	Exterior wall surface m²	Height m	Volume m³
Office 1	60	51	3	180
Kitchen	15	15	3	45
Office 2	80	78	3	240
Corridor	30	12	3	90
Bathroom	15	24	3	45

Table 2: Zone Characteristics

The occupancy density is calculated based on the ASHRAE 62.1 guidelines for office spaces with a value of $10\ m^2$ per person.

In the simulation is assumed a standard working week schedule typical for administrative offices:

- Working Days (Mon-Fri): Occupied from 08:00 to 17:00 (9 hours).
- Weekends: Almost unoccupied

Envelope Thermal Properties and Construction Data

The envelope of the building is entirely based on the reference document *D'Agostino et al., 2019,* where it is mentioned that an isolation thickness over **2-4 cm** is the optimal range.

Wall	U coefficient W/m²K	C J/m²K	theta	phi
Exterior wall	0,611	300600	0,1	0,46
Interior wall	0,526	120000	0,5	0,46
Floor	0,63	664182	0,2	0,46
Ceiling	0,49	629409	0,3	0,46

Table 3: Thermal characteristics with 4 cm isolation thickness

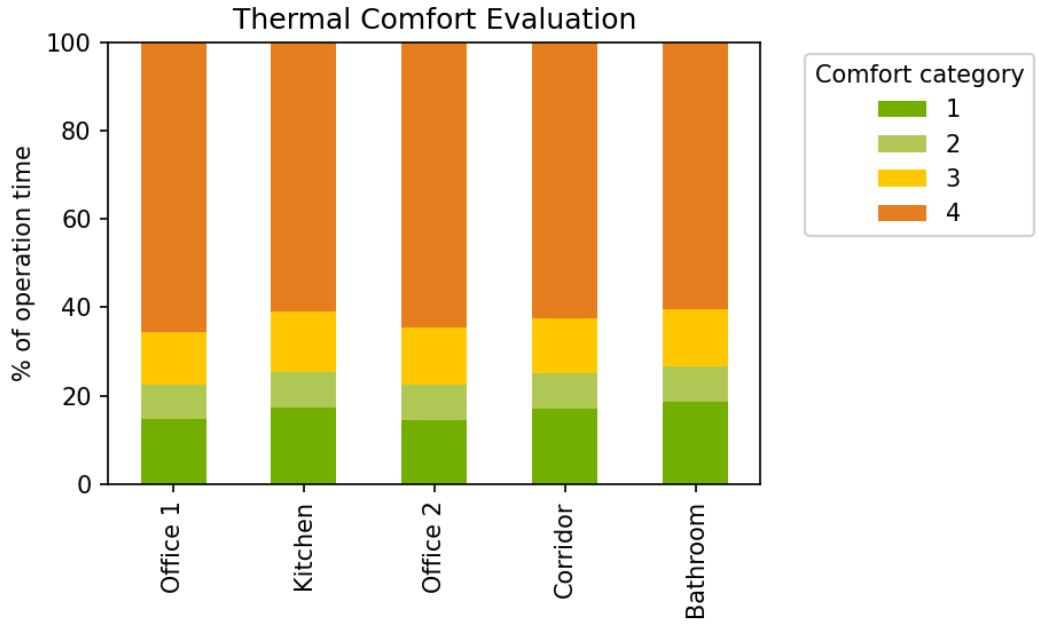
The chosen U-value for the windows is a standard one: it limits heat loss in winter while avoiding excessive heat gain in summer. The windows have a shading factor of 0.69, which means that approximately 69% of solar radiation passes through.

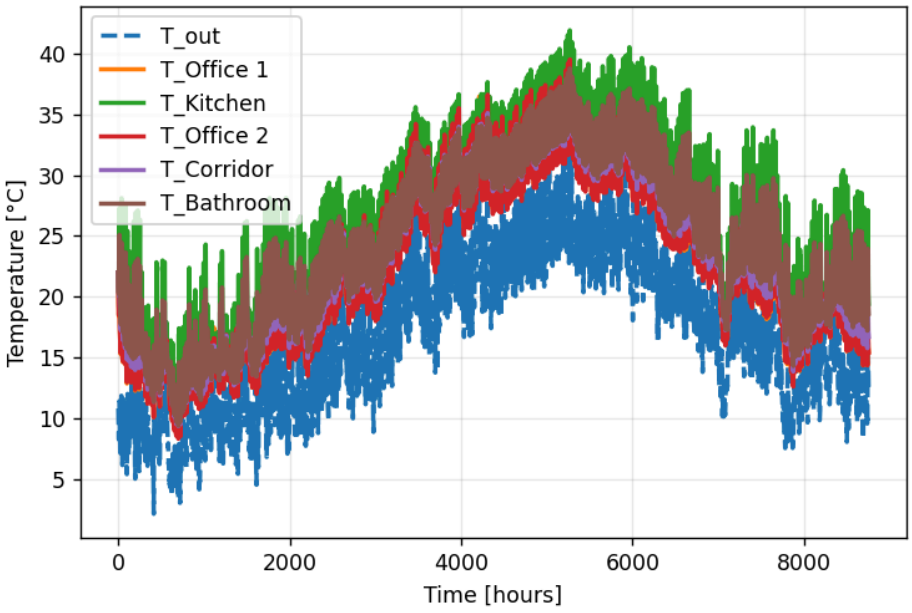
Zone	Area m²	Open area m²	Height m	Orientation	U W/m²K	SF
Office 1	12	2,4	1,15	North	2,05	0,69
Kitchen	3	0,6	1,15	South	2,05	0,69
Office 2	10	2	1,15	West	2,05	0,69
Corridor	2,5	0,5	1,15	South	2,05	0,69
Bathroom	2,5	0,5	1,15	South	2,05	0,69

Table 4: Windows Characteristics

Load Calculation & Sizing

The building model was simulated in passive mode where $T_{supply} = T_{out}$ and $Q_{heating} = Q_{cooling} = 0$ to determine the initial thermal properties. The outcomes of this simulation are shown in the following figures:





As seen, the discomfort category 4 is dominant in the Thermal Control Evaluation graph and temperatures up to **45°C** are reached in the interior of the building, which strongly indicates that a cooling system will be compulsory to reach the comfort zones during the summer. The building will also need heating during the winter due to low temperatures ranging between **10-15°C**, which may cause discomfort.

Design Conditions

The reference document recommends set points of 26ºC (summer maximum) and 20ºC (winter minimum). However, considering the ideal office temperature is 20-22ºC, the operational comfort range is established as **20-24ºC**. Consequently, the heating/cooling system will activate whenever the building temperature is outside this range.

HVAC System Description: Air Handling Unit (AHU)

AHU OPERATION

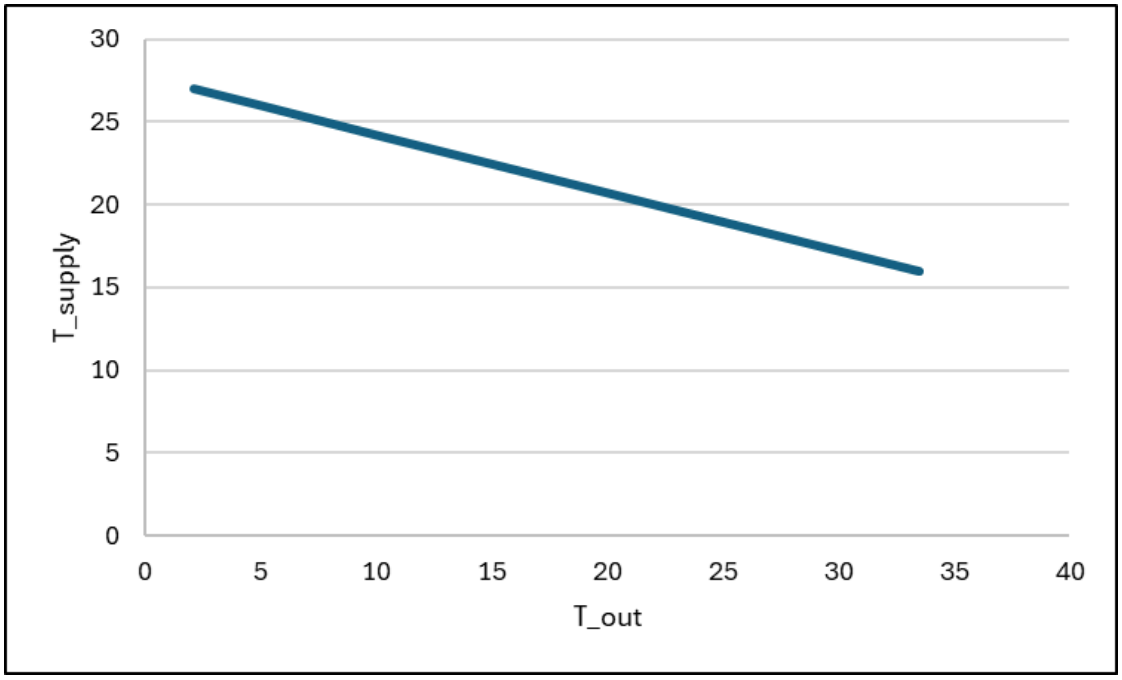
4 logic steps are followed in the AHU to treat the air from T_out to T_{supply}.

Setpoint Calculation

First, the supply air conditions within standard comfort limits are defined as $T_{supply} \in [16,\ 27]\ ºC$ and $w_{supply} \in [0.005,\ 0.012]\ kg/kg$. Operating outside these ranges is not allowed to prevent occupant discomfort.

Within these limits, the specific supply temperature is determined using a weather compensation. This means an inverse relationship with the outdoor temperature: lower outdoor temperatures trigger a higher supply temperature, and vice versa. This behavior is modeled in the code using the following linear equation:

$$T_supply = T_su_max + m_slope * (T_out_t - T_out_min)$$



Where:

- $T_su_max = 27ºC$

- T_out_t is the current outside temperature
- $T_out_max\ and\ T_out_min$ will depend on the climate year
- $m_{slope} = (T_su_min - T_su_max)/(T_out_max - T_out_min)$

Although the supply temperature is determined by the outdoor weather, priority is given to indoor comfort. So , if the indoor temperature exceeds 24ºC, the weather compensation logic is overridden and the supply temperature is fixed at 16ºC to provide immediate cooling.

In terms of humidity, the supply calculation is much simpler:

- If the outdoor air is too dry (< 0.005 kg/kg), the system humidifies it to reach the minimum limit.
- If the air is too humid (> 0.012 kg/kg), it is dehumidified down to the maximum limit.
- If the outdoor humidity is inside this range, the air is supplied as is, without additional humidification or dehumidification.

Fan system

The system will ensure adequate air quality by maintaining a constant hygienic ventilation flow rate of 3,6 $m^3/h/m^2$. The air will be propelled inside the system by a fan. To determine the cost, the electrical power consumption is calculated based on the volumetric airflow and the system's resistance:

1. Volumetric Flow Rate $\dot{V}$: The mass flow rate is converted to volume using the air density :

$$\dot{V} = \dot{m}_{vent} / \rho_{air}$$

2. Calculation of SFP: The total SFP is estimated at 1200 W/(m³/s), composed of the following load contributions:
    - Base Fan Efficiency: 600 W/(m³/s) (High-efficiency standard)

- Heat Exchanger drop: 300 W/(m³/s)
- Filter: 300 W/(m³/s)

3. The power W_{fan} is:

$$W_{fan} = \dot{V} \, x \, SFP$$

Heat Exchanger

To minimize energy consumption, the system prioritizes using the thermal energy from the building´s air T_{in} . This allows for pre-heating or pre-cooling of the incoming outdoor air before any active heating or cooling is applied. The resulting temperature after this recovery process is calculated as $T_{after\ HEX}$ based on the required thermal efficiency ε_{needed}

Operational Mode	Logic Condition	Action
Bypass	$\varepsilon_{needed} < 0$	Disable HEX
Modulation	$0 < \varepsilon_{needed} < 0.7$	Adjust Efficiency
Maximum Capacity	$0.7 < \varepsilon_{needed}$	Fix the Efficiency at 70%

Table 5: Control logic for the Heat Recovery System based on required efficiency.

Heating and Cooling Coils

After the HEX, air enters the coils. The simulation calculates the specific energy required to reach the final supply conditions (T_{supply}, w_{supply}). There are two operational modes:

1. Dehumidification mode: when the humidity of the air taken from the outside exceeds the humidity limit, the system will execute the following procedure:
    - First, the cooling coil lowers the air temperature to condense the excess moisture. A safety limit of 4°C is fixed to prevent the coil from freezing.
    - Consequently, reheating is applied and the heating coil is immediately activated to raise the temperature back to the desired temperature.
2. Standard control mode: If dehumidification is not needed, the system will calculate the net enthalpy difference to determine the necessary operation of heating or cooling.

The necessary energy used by the AHU system is saved in variables to calculate afterwards the dimension of the required heat pump system which will have to supply the energy. The variables are $Q_{heating\ coil}$, $Q_{cooling\ coil}$ and W_{fan}.

TERMINAL UNITS

After studying the behaviour of the building , it is evident that the Air Handling Unit (AHU) alone cannot address the neccesary sensible loads of each zone. The required thermal output Q_{TU} for these units is calculated via a dynamic energy balance which takes into account transmission losses (Walls/Windows), solar gains, infiltration, and internal gains (Occupancy and Equipment).

The zones are equipped with Fan Coil Units and as mentioned a 4-pipe circuit is used, allowing for independent availability of heating and cooling fluids throughout the system.

Control Logic: Proportional Control Strategy

The Terminal Units are regulated via P-control, and three different regimes are found:

The control logic sets up a deadband between 21°C and 24°C ,where the system remains inactive because the temperature is considered within the comfort zone. Outside this interval, a proportional band of ΔT=3°C is applied. Within this band, the heating or cooling output is calculated proportionally to the temperature deviation. If the temperature exceeds the limits of the proportional band , the unit operates in saturation, delivering its nominal maximum capacity P_{nom} to compensate for the peak load.

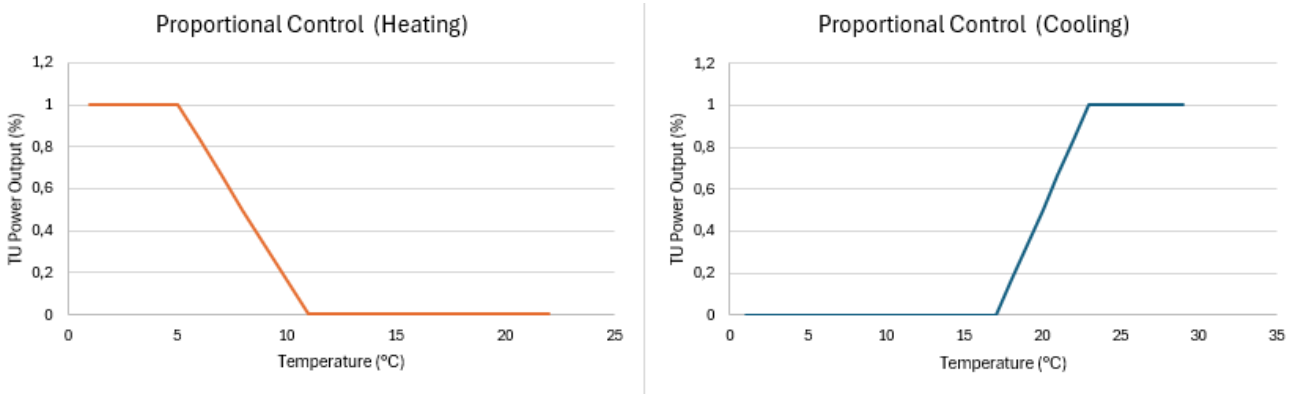


Figure 4: Terminal Unit actuation signal. The heating and cooling outputs are modulated linearly based on the deviation from the setpoints

HEAT PUMP

Dimension of Heat Pump

Based on the hour by hour annual simulation, the building shows a Peak Thermal Demand of ~7.5 kW. Consequently, a HP with a nominal capacity of **10 kW** has been selected. This sizing provides a safety margin of approximately **25%**. This reserve is necessary to ensure thermal comfort during extreme weather situations and to compensate for potential efficiency drops.

System Description and Selection

For the generation system a reversible heat pump is going to be used to run it .It will work as a chiller in summer and will heat during the winter. This unit is capable of reversing its thermodynamic cycle to adapt to the seasonal needs:

- In summer (Chiller Mode): The cycle reverses to produce chilled water at **7°C**, dissipating heat to the outside air.
- In winter (Heating Mode): The unit produces hot water at **45°C** for the heating coils and terminal units.

This technology has been chosen due to economic and operational reasons:

1. **Cost-effectiveness:** Installing a separate chiller and a gas boiler would be too expensive for a relatively small power demand of 10 kW.
2. **Operational Efficiency:** Using only a chiller with electric heaters for winter would be inefficient because resistance heating is 3 times more expensive to operate (COP=1) compared to a heat pump (COP ≈ 3 or 4).

Efficiency Performance

In terms of efficiency, the reference project establishes baseline values of **SCOP 2.8** .In the simulation, a **SCOP of 3.18 i**s obtained in the base scenario. This value is significantly higher than the reference value and it indicates that the system is efficient. This high efficiency

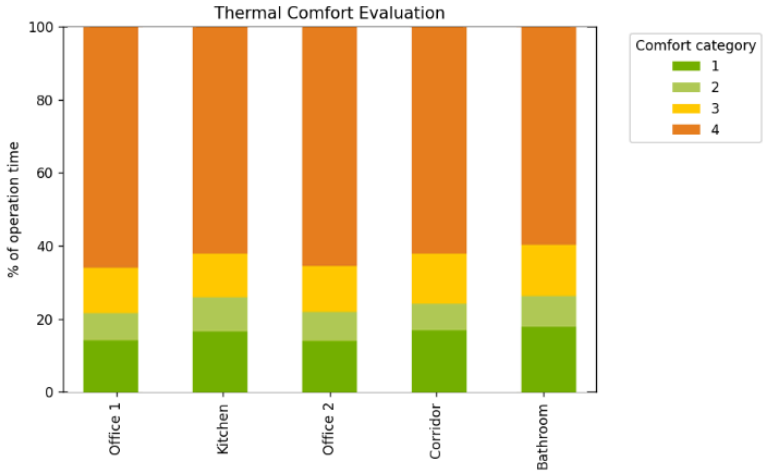
may be driven by the favorable climatic conditions of Palermo, which allow the heat pump to operate frequently at partial loads where efficiency is maximized.

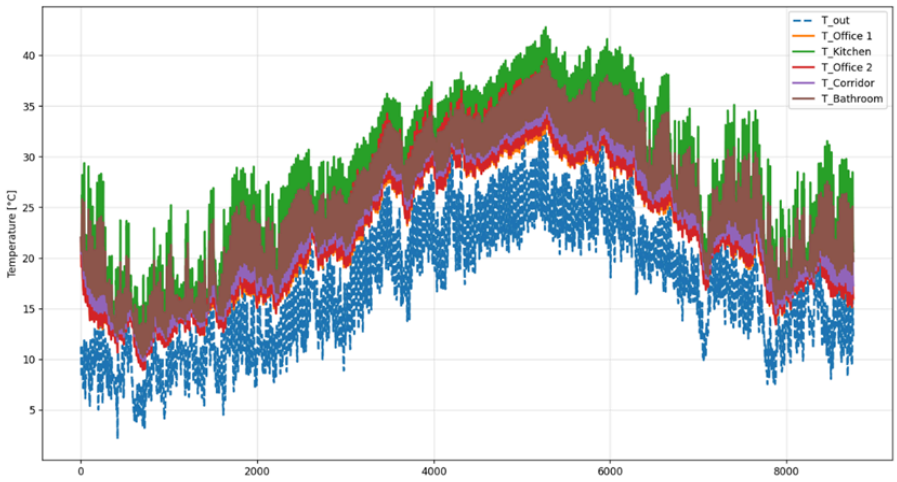
DIFFERENT SCENARIOS

For the evaluation of the building's energy performance and the identification of potential savings, some operational and constructive modifications have been simulated. The following scenarios analyze the impact of key parameters on the final energy consumption and thermal comfort.

No HVAC system

To establish a starting point, a first simulation was run without any active cooling or heating. This allows us to check the building's natural response to passive drivers like infiltration and radiation before sizing the equipment. The results are shown below:

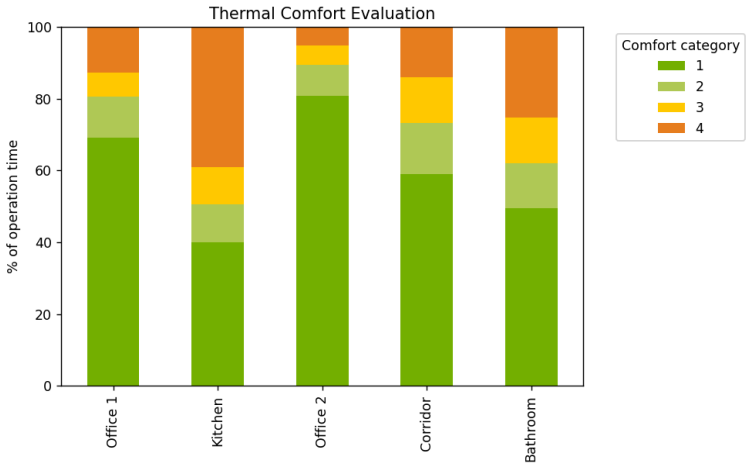


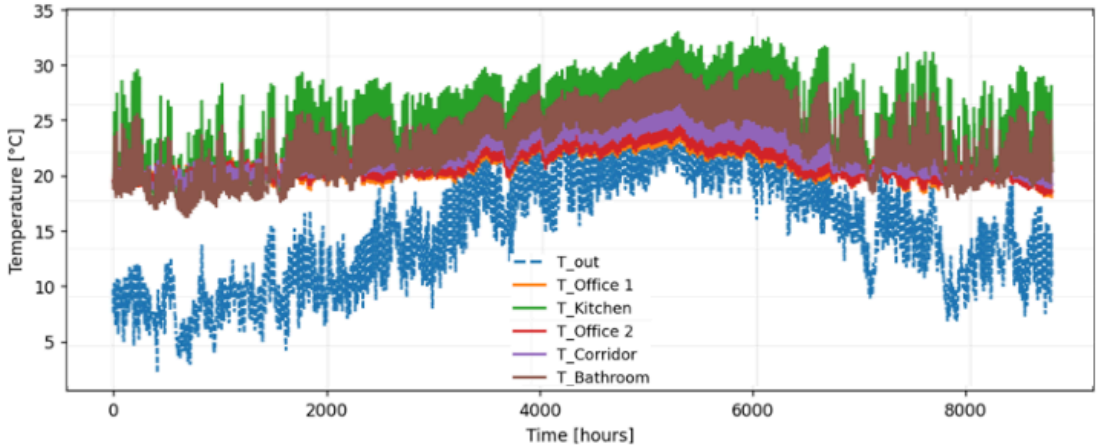


The simulation confirms high levels of overheating. Summer temperatures reach the peak at 45°C, actually exceeding the outdoor temperature which indicates a strong heat retention inside the building. Winter is equally critical with lows of 9°C, proving that the current envelope is insufficient and active intervention is mandatory for both seasons.

Only with AHU

The terminal units are not activated, and the building is conditioned exclusively by the AHU. This approach demonstrates that hygienic ventilation flow rates alone are insufficient to manage the total sensible thermal loads, however, they still contribute to improved indoor comfort and are effective in controlling indoor humidity levels:

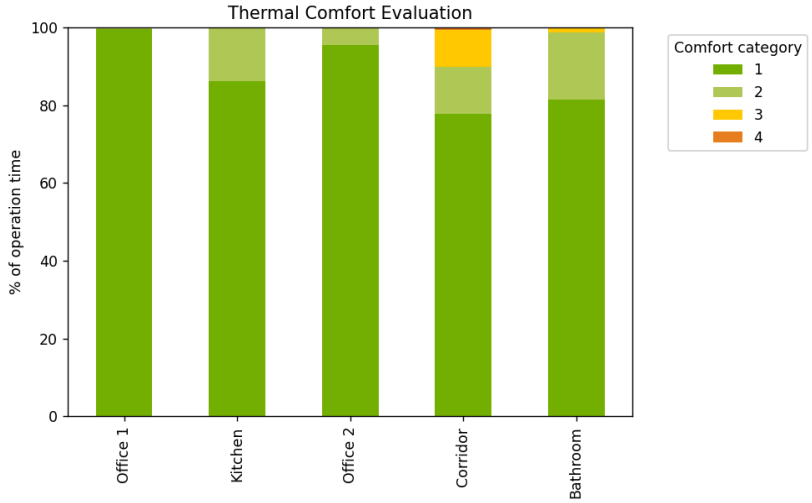


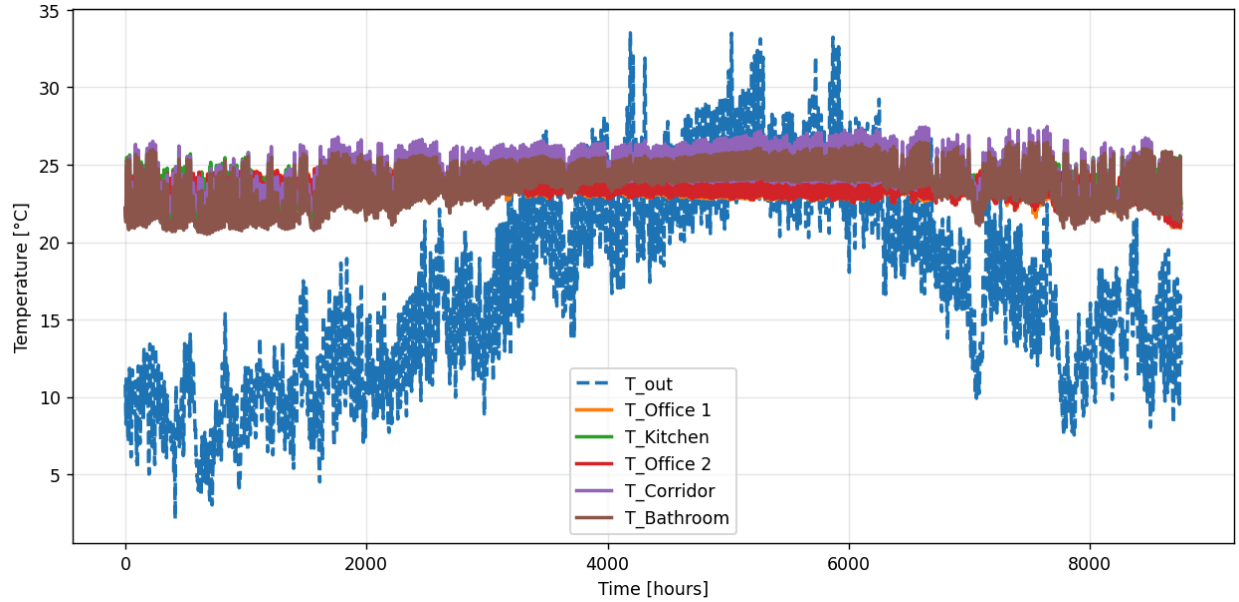


Compared to the first scenario, the maximum indoor temperature is reduced from 45 °C to 35 °C, and the overall temperature profile becomes more balanced, remaining closer to the setpoint values.

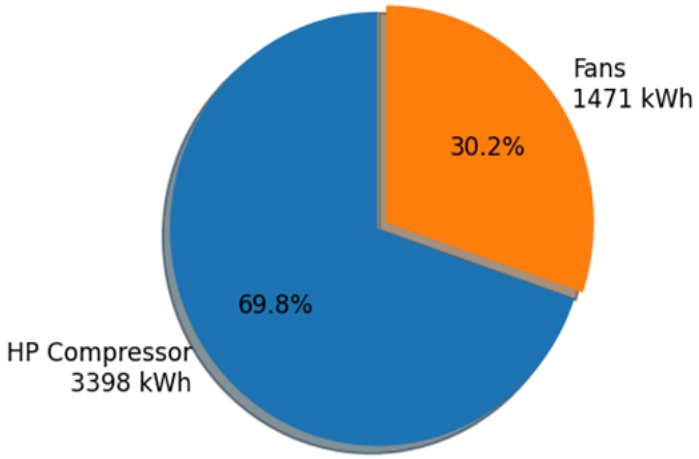
All systems activated during the entire year (BASE scenario)

Below, the baseline scenario is shown where all systems operate continuously without schedule restrictions and terminal units added, representing the worst case consumption.





This graph demonstrates the effectiveness of the complete HVAC configuration, showing that unlike the “Only AHU” scenario, the inclusion of the TU fan coils enables precise sensible load management, with indoor temperatures remaining within the comfort zone for almost all hours as the terminal units remove excess heat in winter and provide the necessary cooling capacity during summer peaks. That confirms that a 10 kW heat pump is the correct size to maintain continuous thermal stability.



As shown, achieving the required comfort conditions results in a total annual energy consumption of 4869 kWh*e*, of which approximately 30% is attributed to the AHU fan operation, while the remaining 70% corresponds to the heat pump compressor. The total annual operating cost is **1704,15€** , based on an energy italian average price of €0.35/kWh.

Switch off during unoccupied hours

A system shutdown (AHU + Terminal Units) during unoccupied hours to maximize savings, is giving the following results:

With the system shutdown during the night, it is observed that the implementation of the intermittent schedule does not significantly alter the comfort levels during occupied hours. The application of the occupancy-based control reduced the annual operational expenditure to **€1201.51**. This represents a **30%** cost reduction. This efficiency is achieved by eliminating energy consumption during unoccupied periods (nights and weekends), thereby maximizing savings without compromising the indoor environmental quality for the users.**Thermal comfort remains within acceptable limits despite the system being switched off during unoccupied hours.**

Changing the Heat Pump for electrical resistances

To quantify the economic and energetic advantages of the HP, the simulation of the base scenario was modified to replace the thermodynamic generation system with a direct electric conversion system. Unlike the HP, which utilizes a compression cycle to transfer heat from the environment (efficiencies of more than 100%), the electric resistance system operates on the Joule effect. This results in a fixed Coefficient of Performance (COP) and Energy Efficiency Ratio (EER) of exactly 1:

Metric	Heat Pump	Elec. Resistance	Difference
Average Efficiency	3-4	1	**-70%** Efficiency
HVAC Consumption	3409 kWh_e	15478 kWh_e	**+354%** energy cons.
HVAC Cost	~1193 €	5417 €	**+4224** €
Fan Cost	~511 €	686 €	**+175** €
TOTAL ANNUAL BILL	~1704 €	6104 €	**3.6x** more expensive

The simulation confirms that using direct electric heating/cooling (resistances) is economically unsustainable for this building type and climate. The HP is essential, as it takes ambient energy to provide approximately 70% of the required thermal load for "free". Reverting to a system with *COP* = 1 would entail an extra cost of over **4000€ per year**, which makes no sense for an office building of 200 m^2.

Change the insulation of the walls (4cm to 2cm)

This scenario tests the building envelope's relevance. The thickness of the thermal insulation layer is reduced from 4 cm to 2 cm, increasing the U-value. The results are shown and compared with the BASE scenario below:

Metric	4 cm Insulation (BS)	2 cm Insulation	Difference
Heating Demand	3371 kWh*t*	3845 kWh*t*	**+474** kWh*t*
Cooling Demand	12106 kWh*t*	11048 kWh*t*	**-1058** kWh*t*
Total Annual Bill	1704 €	1687 €	Similar

When increasing the thermal transmittance **U,** higher heat losses in winter appears . As shown in the table, heating demand rises by 14%. However, this thinner insulation proves beneficial during the cooling season in summer because by reducing the thickness to 2 cm, the building envelope allows for better dissipation of internal heat loads, resulting in a significant drop in cooling demand.

The results indicate a thermal equilibrium: the extra energy required for heating is fully compensated by the energy savings in cooling.

These findings corroborate the research by D. D'Agostino et al. which suggests that for this specific climatic zone (Palermo) and building typology, the optimal cost insulation thickness lies between the 2–4 cm range. Exceeding this thickness yields diminishing returns or negative impacts due to cooling penalties.

From an economic perspective, **2 cm** is the best option. It offers a lower annual operational cost and a lower initial capital investment, as thinner insulation material is less expensive.

Install blinds in the building (Solar Factor 0.69 to 0.1)

In summer, Palermo experiences high cooling loads, which is why this scenario introduces the use of external shading devices like blinds. The solar factor of the windows is reduced from 0.69 to 0.1, blocking direct solar radiation and then minimizing solar gains in summer.

The comparison between the scenarios below:

Metric	Solar Factor 0.69 (BS)	Solar Factor 0.1	Difference
Heating Demand	3371 kWh*t*	4936 kWh*t*	**+1565** kWh*t*
Cooling Demand	12106 kWh*t*	5478 kWh*t*	**-6628** kWh*t*
Peak Power Demand	7.6 kW	4.17 kW	**-3.43** kW
Total Annual Bill	1704 €	1361 €	**-343** €

Consequently, installing blinds in Palermo is worth it because the significant cooling savings easily offset the small additional heating cost. In addition, opening the blinds during winter allows solar radiation to enter the building and reducing this heating cost, helping regulate indoor temperatures and maintain energy efficiency throughout the year.

FINAL SCENARIO

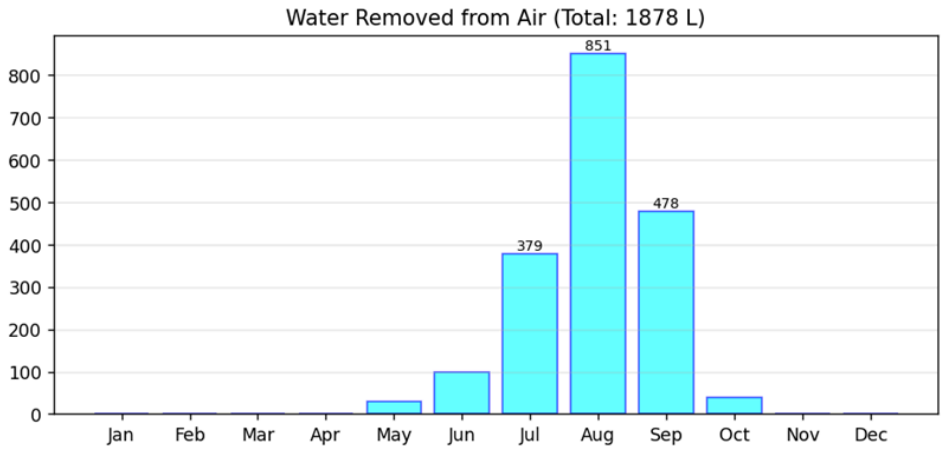
Scenario Definition

This "Final Scenario" represents the most optimized building configuration mixing the best scenarios which have been run before. The key parameters modified are:

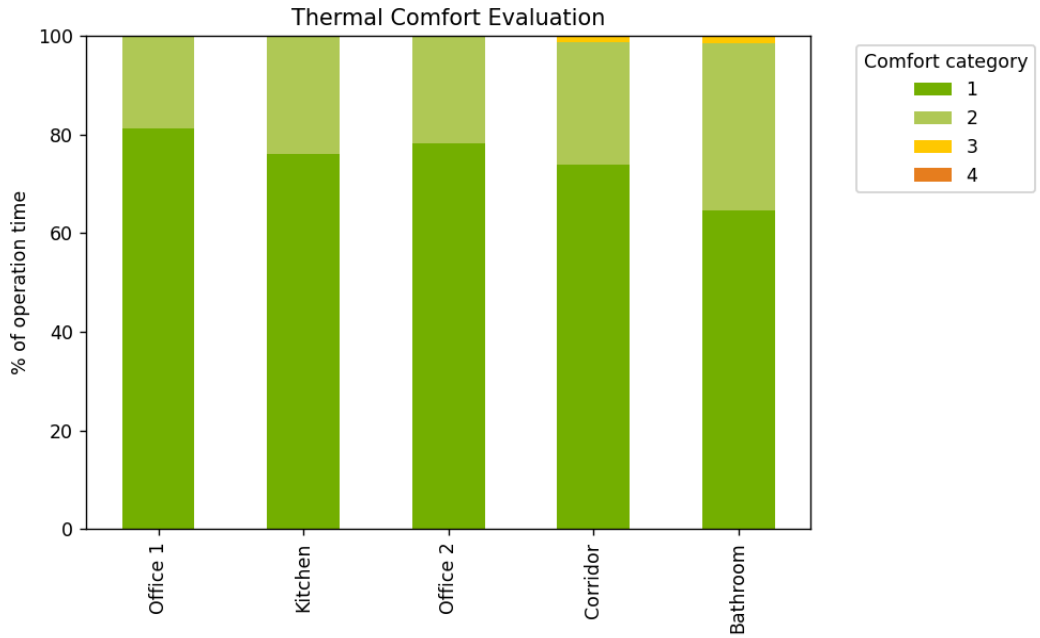
- HVAC Operation: System switched OFF during non-occupied night hours (20:00 - 08:00) to prevent unnecessary cooling of an empty building.
- Envelope: Insulation thickness adjusted to 2 cm (down from 4 cm), optimizing material use based on the reduced thermal load.
- Transparent Envelope: Implementation of external shading devices, reducing the SF from 0.69 to 0.4.
- Generation: Heat Pump system modeled instead of electrical resistances.

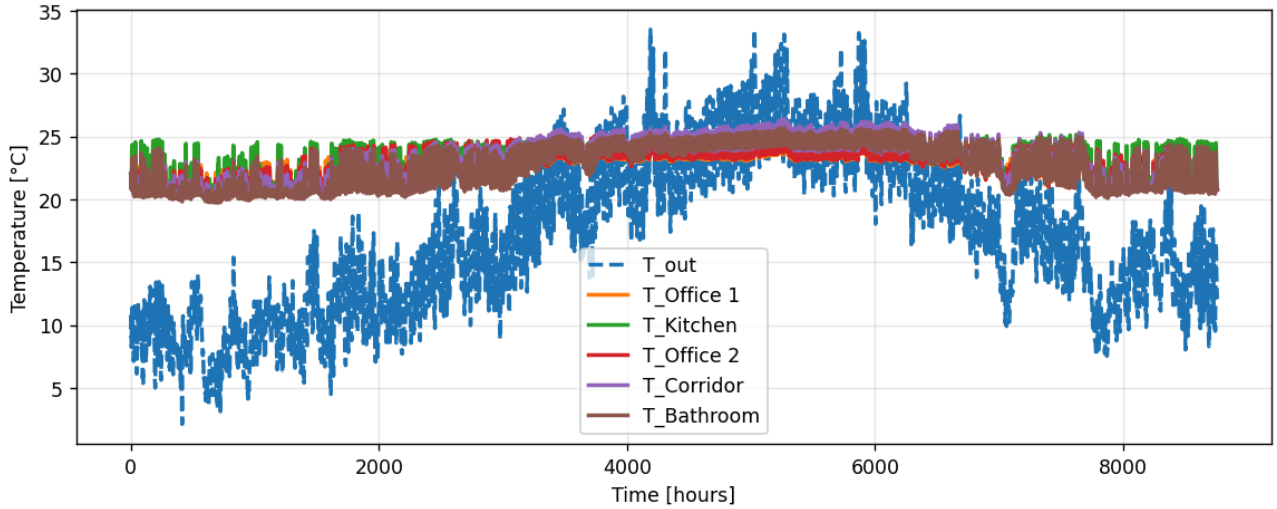
Interpretation of Results

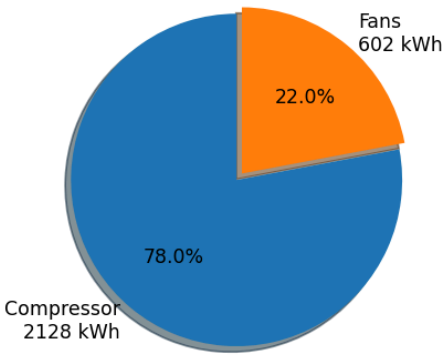
In terms of dehumidification, the total water removed from the building is shown in the next figure:



The maximum quantity of dehumidification is found in **August**, with a **maximum daily peak** of 53.0 liters/day. It makes sense because it is the most humid month according to the TMY data, exhibiting the highest specific humidity ratios of the year.







The total energy consumption of the building is 2729 kWh/year and the total operating cost is **955.41€**. These results represent a high-performance scenario and for a 200 m² office building in Palermo are considered efficient from a thermal comfort

perspective. This value corresponds to approximately 13.65 kWh/m^2 per year, which aligns with typical benchmarks for office buildings designed to high efficiency standards *(Passivhaus standard = 15 kWh/m^2).*

CONCLUSION

The energy efficiency achieved in this final simulation (13.65 kWh/m^2) is not accidental , it is the direct result of stacking the most effective strategies identified in the previous scenarios.

But, while these results demonstrate the theoretical potential of passive design strategies, the energy consumption should be interpreted as an **idealized scenario**. Several factors contribute to this optimism and must be considered for a real-world application.

First of all, maintaining a constant SF of 0.4 represents an aggressive strategy. This value assumes the correct use of external blinds to block 60% of solar radiation. In a realistic scenario, occupants typically open and close the blinds based on their visual needs or preference for natural light. If users leave blinds open during peak solar hours, the cooling load will increase significantly.

Furthermore, the system efficiency is idealized in some aspects. The simulation focuses on zone demand and generation, effectively ignoring thermal losses in the distribution network. The heat loss in pipes and ducts, air leakage, or the parasitic energy of pumps and valves are not modeled here, yet they typically add around 10-15% to the total building energy use.

In conclusion, this final scenario confirms that combining solar control, with an optimized schedule and using a HP as the heat generation source is the single most effective strategy for Palermo. However, it should be noted that real-world consumption will likely be higher.